



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 – 23



**CTD Iron Rice Mixer
Model: 2009**

CTD 2009 IRON RICE MIXER

Test Report No.	:	2013 - 23
Date of Test	:	March 22, 2013
Test Valid Until	:	March 22, 2018
Brand	:	CTD
Model	:	2009
Type	:	Gravity-type, manually-operated
Manufacturer	:	FREDVAR ENTERPRISE Sogod - Tibgao Virac, Catanduanes
Test Requested By	:	Manufacturer
Machine Classification	:	Commercial

SUMMARY

The CTD 2009 Iron Rice Mixer was tested using 500 kg of milled rice and 2.5 kg of fortificants (iron). By means of gravitational force and the mixing chamber, milled rice and fortificants (iron) were mixed inside the equipment to produce Iron Fortified Rice (IFR).

Results of the performance tests showed that the CTD 2009 Iron Rice Mixer had an Output Capacity of 5657 kg/h. During the mixing process, twelve persons were needed. These were the following: two bag opener and tosser (get the sack of rice on the pile and tosses to another laborer at the feeder); four feeders (carry the opened bag of rice and pour it to the hopper of the mixer); one operator (responsible for pouring the fortificant to its hopper); two baggers (bag the IFR produced and toss

to piler for proper piling); two pilers (pile the sacks of IFR); and one checker (monitors dispersion analysis).

Laboratory analysis of the samples from the ten sacks of Iron Fortified Rice (IFR) randomly taken at the top, middle and bottom of each sack showed that the percentage of fortificants present in the product was 0.47% with percent variation of $\pm 0.51\%$.

METHOD AND SCOPE OF TEST

The iron rice mixer was tested using 500 kg of milled rice and 2.5 kg of fortificants (iron). Ten sacks of Iron Fortified Rice were produced. The output capacity of the equipment to produce IFR was recorded. Samples were randomly taken from each sack of IFR with two samples each from the top, middle and bottom of the sack.

Laboratory analysis of the samples taken from Iron Fortified Rice (IFR) was conducted to determine the actual percentage of the fortificants present in the product.

PLACE AND DATE OF TEST

The iron rice mixer was tested at the National Food Authority (NFA) Warehouse, San Isidro Village, Virac, Catanduanes on March 22, 2013.

DESCRIPTION OF THE EQUIPMENT

The CTD 2009 Iron Rice Mixer as shown in **Figure 1** was a gravity-type, manually-operated equipment for mixing milled rice with fortificants (iron). It was made of light material, particularly good lumber, formica, and plywood. It was made of three sections: the hopper-receiver, the main mixer and the main stand which also served as the bagging section. The equipment was collapsible into these three sections. It had an overall dimensions (L x W x H) of 1305 mm x 1060 mm x 2590 mm.

The hopper-receiver which was at the top layer of the equipment consisted of the hopper for milled rice (**Fig. 2**) which was lined with formica and the hopper for the fortificants (**Fig. 3**). Installed at the delivery tube of the hopper for the fortificant was a 36 watts electric motor with blower (**Fig. 4**) to serve as vibrator to avoid clogging of the fortificants inside the delivery tube.

The second or middle layer was the mixing chamber with two types of mixing mechanism. The upper mixing mechanism consisted of triangular cut wood (**Fig. 5**) alternately arranged consisting of six layers which served as the main mixer or splitter. The lower mixing mechanism consisted of a propeller-type (**Fig. 6**) mixing mechanism with four blades which provided further mixing of milled rice and fortificants.

The lower layer which served as the main stand of the equipment was where the output chute (**Fig. 7**) was installed.



Figure 1. CTD Iron Rice Mixer
Model: 2009



Figure 2. Hopper for milled rice



Figure 3. Hopper for fortificants



Figure 4. Electric motor with blower

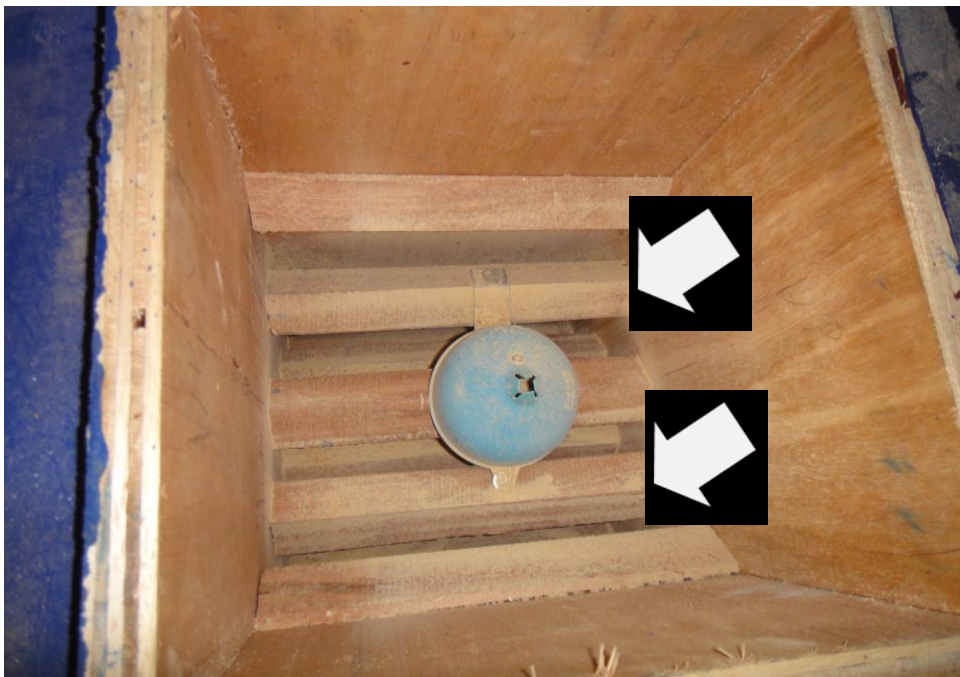


Figure 5. Main mixer or splitter



Figure 6. Propeller-type mixer



Figure 7. Output chute

RESULTS AND DISCUSSION

The test was conducted by using 500 kg of milled rice and 2.5 kg of fortificants (iron). By means of gravitational force and the mixing mechanisms of the equipment, Iron Fortified Rice (IFR) was produced. The fortification was in line with RA 8976 Food Fortification Act of 2000 which provided for mandatory fortification of staples like rice, sugar, oil and flour in the Philippines by November 7, 2004.

Results of the performance tests showed that the CTD 2009 Iron Rice Mixer had an Output Capacity of 5657 kg/h.

Laboratory analysis of the samples from the ten sacks of Iron Fortified Rice (IFR) randomly taken at the top, middle and bottom of each sack showed that the percentage of fortificants present in the product was 0.47% with percent variation of $\pm 0.51\%$.

Results of tests are shown in **Table 1** and **Table 2**.

Labor Requirements

During the mixing process, twelve persons (**Fig. 8**) were needed. These were the following: two bag openers and tossers (get the sack of rice on the pile and tosses to another laborer at the feeder); four feeders (carry the opened bag of rice and pour it to the hopper of the mixer); one operator (responsible for pouring the fortificant to its hopper); two baggers (bag the IFR produced and toss to piler for proper piling); two pilers (pile the sacks of IFR); and one checker (monitors dispersion analysis).

Observations

- 1) The iron rice mixer had the capability to produce iron fortified rice (**Figure 9**) by addition of fortificant (**Figure 10**) at the desired quantity.
- 2) The CTD Iron Rice Mixer was collapsible (**Figure 11**) for ease of transport.



Figure 8. Operators of the mixer

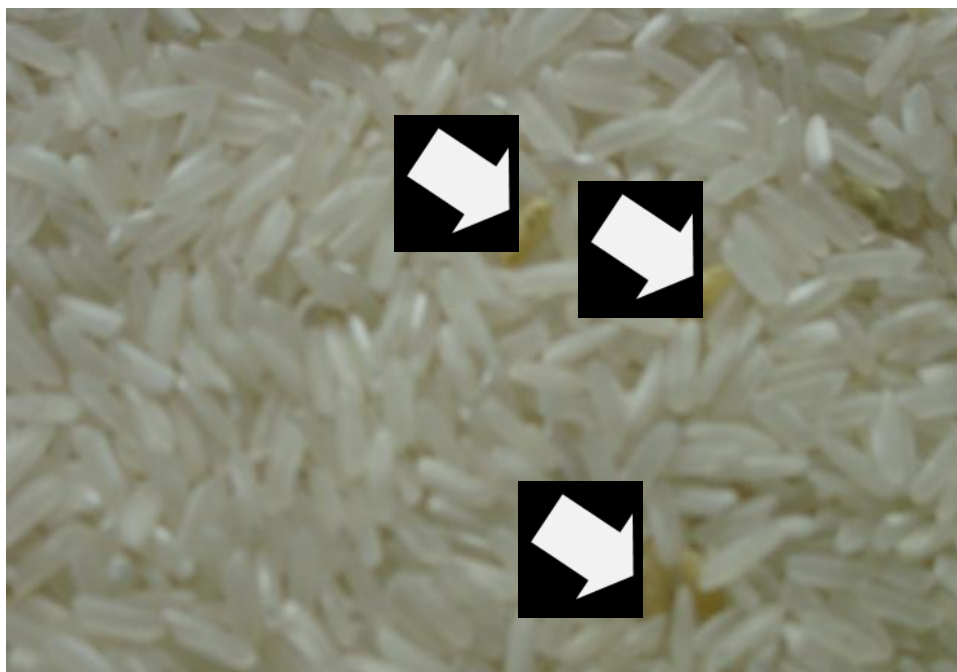


Figure 9. Iron Fortified Rice (IFR)

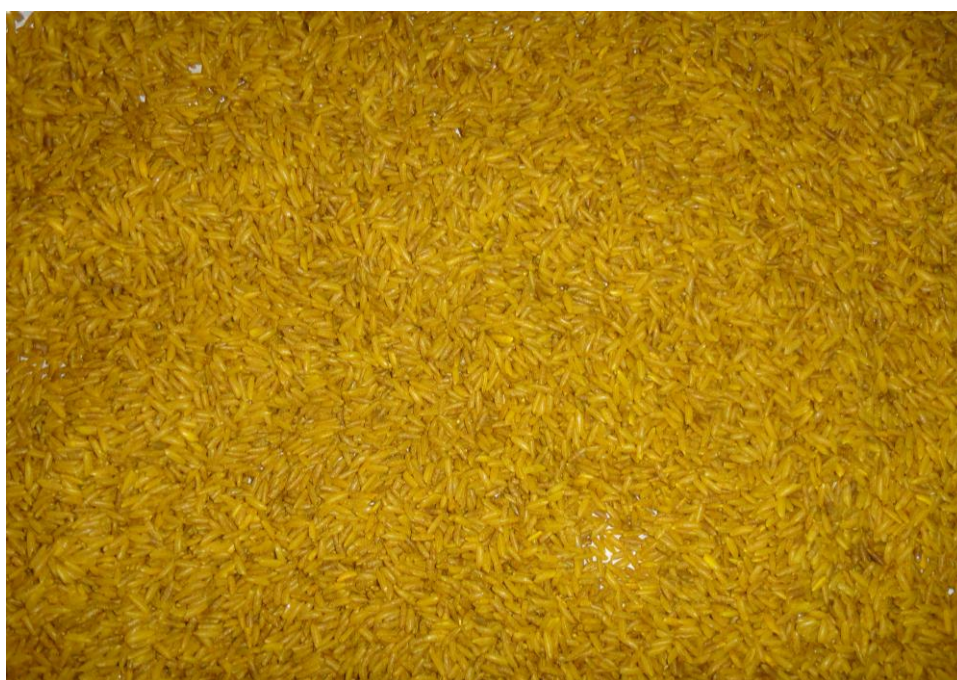


Figure 10. Fortificants



Figure 11. The mixer was collapsible

PERFORMANCE TEST

Machine Tested : CTD Iron Rice Mixer
Model: 2009

Location of Test: NFA Warehouse
San Isidro Village
Virac, Catanduanes

Date of Test : March 22, 2013

Table 1. Results of Performance Test

ITEM	DATA
1. Weight of milled rice (kg)	500
2. Weight of fortificants (kg)	2.5
3. Total time of operation (min)	5.33
4. Output capacity (kg/h)	5657

Table 2. Laboratory Analysis of Iron Fortified Rice (IFR)

Item	Amt, of Sample (g)	Amt. of Milled Rice (g)	Amt. of Fortificant (g)	Percent Fortificant
Sample 1	1000	995.30	4.70	0.47
Sample 2	1000	995.34	4.66	0.47
Sample 3	1000	995.50	4.50	0.45
Sample 4	1000	996.12	3.88	0.39
Sample 5	1000	995.10	4.90	0.49
Sample 6	1000	994.55	5.45	0.54
Average	1000	995.32	4.68	0.47
Variation (%)	-	-	±0.51	-

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N.B.

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